EXP 10: Write a Python Program to Implement Expectation & Maximization Algorithm

CODING:

```
import math


# Input Dataset
X = [1, 2, 3, 8, 9, 10]


# Initialize Parameters
means = [2.0, 9.0]
variances = [2.0, 2.0]
weights = [0.5, 0.5]


# Gaussian Probability Function
def gaussian(x, mean, variance):
    return (1 / math.sqrt(2 * math.pi * variance)) * math.exp(
        -((x - mean) ** 2) / (2 * variance)
    )


# EM Algorithm
for iteration in range(20):


    # E-Step: Calculate Responsibilities
    responsibilities = []


    for x in X:
        p1 = weights[0] * gaussian(x, means[0], variances[0])
        p2 = weights[1] * gaussian(x, means[1], variances[1])
```

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```
    total = p1 + p2

    r1 = p1 / total
    r2 = p2 / total

    responsibilities.append([r1, r2])

# M-Step: Update Parameters
for k in range(2):

    total = sum(r[k] for r in responsibilities)

    means[k] = sum(
        responsibilities[i][k] * X[i]
        for i in range(len(X))
    ) / total

    variances[k] = sum(
        responsibilities[i][k] * (X[i] - means[k]) ** 2
        for i in range(len(X))
    ) / total

    variances[k] = max(variances[k], 0.000001)

    weights[k] = total / len(X)

# Display Results
```

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```
print("Input Data:", X)

print("\nFinal Cluster Means:")
print(means)

print("\nFinal Variances:")
print(variances)

print("\nCluster Weights:")
print(weights)

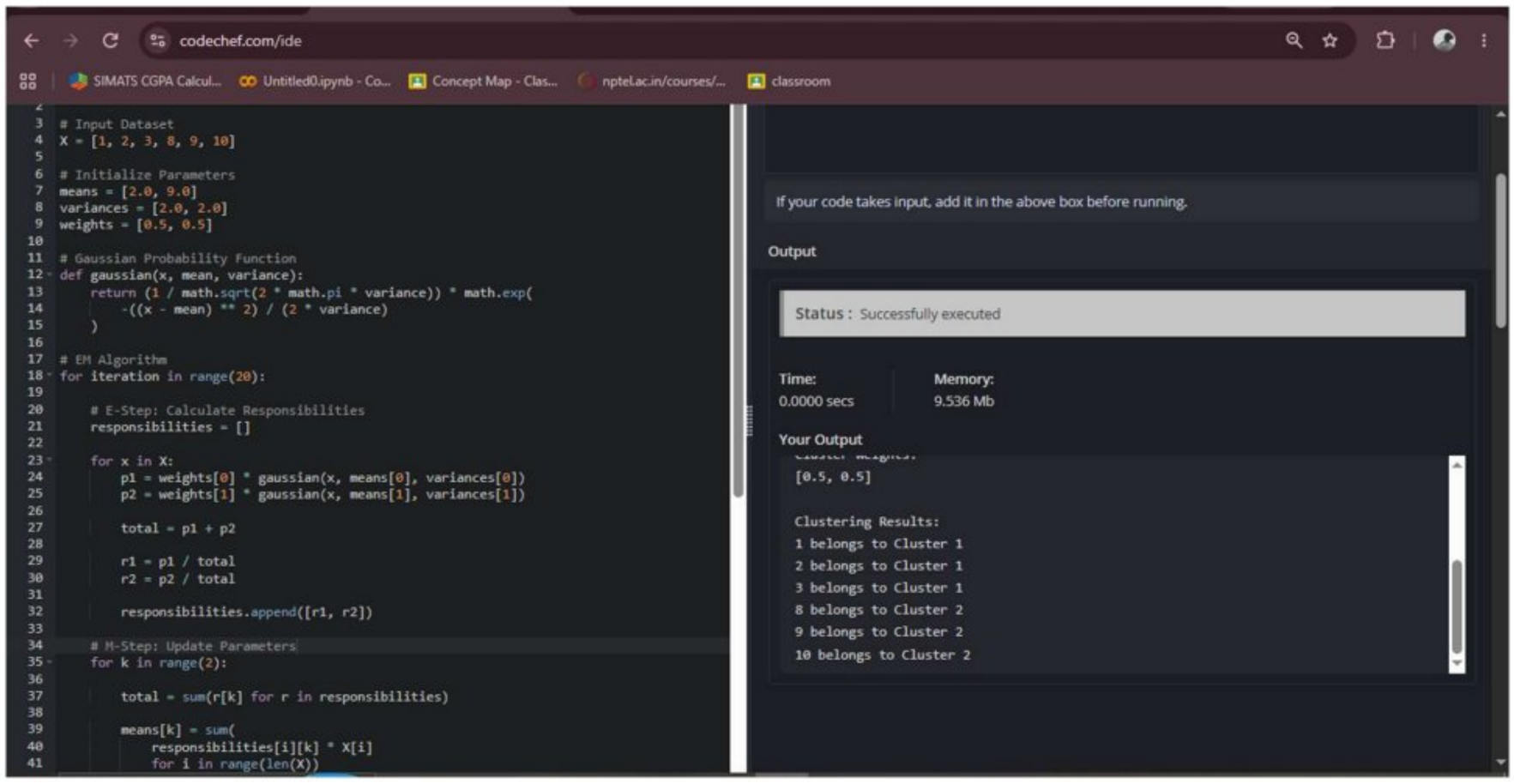
# Assign Clusters
print("\nClustering Results:")

for x in X:
    p1 = weights[0] * gaussian(x, means[0], variances[0])
    p2 = weights[1] * gaussian(x, means[1], variances[1])

    if p1 > p2:
        print(x, "belongs to Cluster 1")
else:
    print(x, "belongs to Cluster 2")
```

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OUTPUT:



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